

Finite Element Methods – APE2

1D FEM

1 Can a finite element solution be nodally exact ?

Consider the Poisson

$$-\frac{d}{dx}\left(\kappa(x)\frac{du}{dx}\right) = f(x) \quad \text{in } \Omega = (0, 1), \quad (1)$$

with the boundary conditions

$$u(0) = 0, \quad u(1) = 0 \quad (2)$$

1. Check with the code that the the finite element solution is nodally exact in the case $\kappa(x) = 1$ and $f(x) = 1$. Is it still exact for $(\kappa(x) = 1+x, f(x) = 1)$, $(\kappa(x) = 1, f(x) = 1+x)$, $(\kappa(x) = 1+x, f(x) = 1+x)$?
2. Prove that 1D finite elements do give nodally exact solution when κ does not depend on x .

2 Finite element convergence

We continue with the choice $\kappa(x) = 1, f(x) = 1$.

1. Compute the exact energy
2. With the code, get the finite element energy.
3. Check that finite element energy is always above the exact one.
4. Analyse the way the energy is converging with the mesh size.

3 A more complex energy

Consider the following energy

$$\Pi(u) = \int_{r_0}^{r_1} r (u')^2 + \frac{1}{4} \frac{u^2}{r} dr \quad (3)$$

with Dirichlet conditions

$$u(r_0) = u_0, \quad u(r_1 = 1) = 1$$

where $0 \leq r_0 \leq 1$.

1. Find the associate finite element problem.
2. Modify the code so that it can analyze this new energy function.
3. Compare the finite element solution to the exact one, and the finite element energy to exact one.
4. What happens if $r_0 = 0$ and how can you handle it in the finite element formulation?